

16	A	As the light passes through the first sheet half of the intensity of the light is lost. The second sheet will need to reduce the intensity by another half in order for the emerging light to be $0.25I_o$. So,

$$\frac{I_o}{2} \cos^2 \theta = \frac{I_o}{4}$$

$$\cos^2 \theta = \frac{1}{2}$$

$$\cos \theta = \sqrt{\frac{1}{2}}$$

$$\theta = \cos^{-1} \sqrt{\frac{1}{2}} = 45^\circ$$

Either sheet could be moved as both will reduce the intensity by half each time the light passes through.